Roll No.

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharmacy (Sem–3)

# PHYSICAL PHARMACEUTICS-I

Subject Code : BP-302T

M.Code : 75106

Date of Examination : 12-06-2023

Time : 3 Hrs.

Max. Marks : 75

**INSTRUCTIONS TO CANDIDATES :**

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

## SECTION-A

1. **Define briefly :**

   a) What do you know about BCS classification?

   b) How solubility plays an important role.

   c) What is crystalline nature?

   d) What are surfactants? give examples.

   e) Define the liquid interface.

   f) What is dissociation constant?

   g) Discuss the scale used for polymers.

   h) Define interfacial tension.

   i) What is protein binding?

   j) Describe the role of physical pharmaceutical studies.

## SECTION-B

2. Give a note on factors affecting solubility.
3. Write complete note on the inhaler with a diagram.
4. Write a comment method to identify surface interfacial tension.

## SECTION-C

5. Give a note on the eutectic mixture.
6. Discuss the diffusion principles in biological systems.
7. What are the various roles of buffers in the pharma sector?
8. Elaborate on the applications of distribution law.
9. Diacuss the various necessary aspects for stability.
10. Comment on adsorption at liquid interfaces.
11. Justify the buffered equation.
12. What is Sorensen scale?
13. Mention buffer capacity.

**NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.**



July - 2023